List of Errata to

Introduction to Multiple Antenna Communications and Reconfigurable Surfaces

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This document lists typos detected in the published manuscript of:

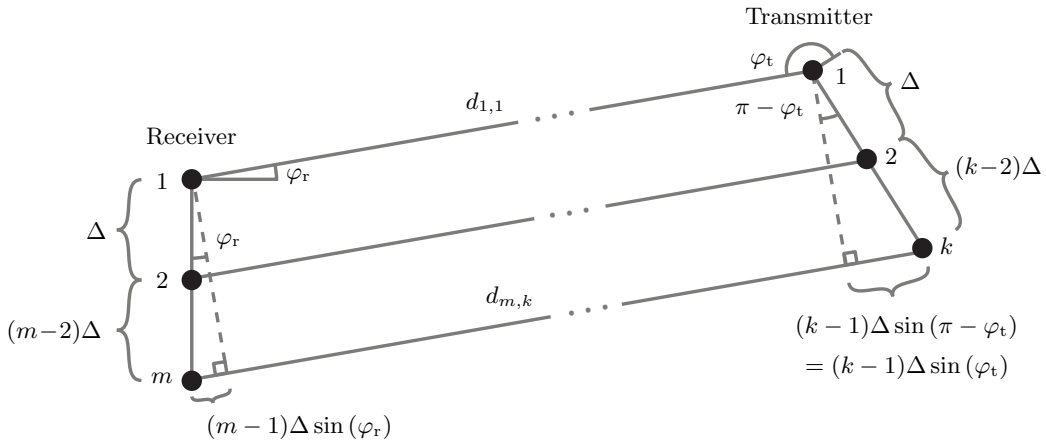
Emil Björnson, Özlem Tuğfe Demir (2024), "Introduction to Multiple Antenna Communications and Reconfigurable Surfaces", Boston-Delft: now publishers, http://dx.doi.org/10.1561/9781638283157

This document will be updated if further typos are discovered.

Please report potential typos to us so we can verify it and then update this file. You can do this by sending an email to emilbjo@kth.se

List of Errata

- In the first sentence on page 191, "(3.102)" in "The transmitted signal $\mathbf{x}$ in (3.102)" should be "(3.101)".
- In the third paragraph of Example 4.4, "$\log_2(81) = 6.34$" should be "$\log_2(81) \approx 6.34$".
- In Example 4.5, the sentence "The ML estimate of α is obtained by modifying (4.43) as" should be "The ML estimate of α is obtained by maximizing the exponent in (4.39) in terms of α as" . Moreover, the sentence "We notice that $\hat{\alpha}$ is the square root of the ML estimate of β in (4.43) (when $\hat{\varphi} = \varphi$)" in the next paragraph of the same example should be removed.
- In Figure 4.23, the angle-of-departure φ_{t} at the transmitter was measured clockwise instead of counterclockwise as elsewhere in the book. For better consistency, the figure should be replaced by:



- In (4.130), $\mathbf{a}_3(\varphi, \theta)$ should be $\mathbf{a}_3(\theta, 0)$.
- On footnote 2 of page 610, "A capacitor adds a positive value to X_n and an inductor adds a negative value." should be "A capacitor adds a negative value to X_n and an inductor adds a positive value."
- In Eq. (4.95), $\mathbf{a}_K(\varphi_{\mathrm{t}})$ should be replaced by $\mathbf{a}_K^*(\varphi_{\mathrm{t}})$.

- There should be an integral sign in the last expression of Eq. (2.189) as

$$P_{\mathrm{D}} = \int_{\gamma'}^{\infty} f_{\mathrm{y}|\mathcal{H}_1}(y|\mathcal{H}_1)\,\partial y = \int_{\gamma'}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-1)^2}{2\sigma^2}} \partial y$$

  .

- On page (160), above Eq. (3.23), in the sentence "...it can be either used to get a larger SNR, or we can reduce q by a factor $1/\|\mathbf{h}\|^2$ to get the same SNR as in the single-antenna case using less transmit power", $1/\|\mathbf{h}\|^2$ should be replaced by $1/M$. So the sentence becomes "...it can be either used to get a larger SNR, or we can reduce q by a factor $1/M$ to get the same SNR as in the single-antenna case using less transmit power."

- There is a redundant "x" next to Eq. (4.54) in "The general trend is that the function in (4.54)x reduces as the argument attains larger positive or negative values, but it also oscillates and has zero-crossings for integer-valued arguments.' on page 228.

- In the second paragraph of Example 3.9, $\mu - s_2^2/N_0$ and $\mu - s_1^2/N_0$ must be $\mu - N_0/s_2^2$ and $\mu - N_0/s_1^2$, respectively.